

SMART AEROSPACE STRUCTURES

V Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE15	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	INTRODUCTION TO SMART STRUCTURES							
Smart/Intelligent structure, Classification, Key elements of a smart structure, Actuators and Sensors, Control Methodologies and Strategies, Integration Issues, Advantages and Applications								
UNIT-II	INTRODUCTION TO PIEZOELECTRIC MATERIAL							
Microstructure of Materials, Crystal Symmetry, 3D point groups, Forms of Piezoelectric Material, Constitutive Modeling of Piezoelectric Material, Piezoelectric Sensors and Actuators, Numerical problems.								
UNIT-III	STRUCTURAL ANALYSIS							
Induced strain Actuation – Static Analysis, Energy principles for strain analysis, Dynamic analysis of beam - Induced strain actuation using Energy principles, Energy harvesting and Vibration control, Introduction to analysis of composite laminate with piezoelectric patches.								
UNIT-IV	SHAPE MEMORY ALLOYS							
Introduction to shape memory alloys, Temperature and stress dependent phase transformation modeling, Stress strain curve at low temperature, Constitutive relations, Finite element formulation of Euler-Bernoulli beam, Analysis of a beam with shape memory alloy wire.								
UNIT-V	STRUCTURAL HEALTH MONITORING							
introduction - aircraft structural damage - ageing aircraft problem - lifecycle cost of aerospace structures - aircraft structural design - damage monitoring systems in aircraft - non-destructive testing - structural health monitoring - emerging monitoring techniques and sensor technologies-monitor evaluation tests- large-scale Metallic Evaluator - Large-scale Composite Evaluator								
Text Books:								
1.Srinivasan A .V and Michael McFarland, “ Smart Structures: Analysis and Design”, Cambridge University Press , 2001								
2.Walter Brown, “Handbook of Smart Structures and Materials: Properties, Technologies and Applications”, Oxford Book Company.								
.. Reference Books:								
1. Inderjit Chopra and Jayant Sirohi, “Smart Structures Theory: 35”, Cambridge University Press.								
2. Erasm o Carrera, Salvatore Brischetto and Pietro Nali , “Plates and Shells for Smart Structures: Classical and Advanced Theories for Modeling and Analysis, Wiley 1st edition.								
Web References:								
1. https://www.sciencedirect.com/topics/physics-and-astronomy/smart-structures								
2. https://onlinelibrary.wiley.com/doi/10.1002/9780470682401.ch1								
3. https://onlinelibrary.wiley.com/doi/10.1002/9780470682401.ch5								
4. https://link.springer.com/book/10.1007/978-3-319-22413-8								
E-Text Books:								
1. Vinod K Wadhawan, “Smart Structures”, Oxford University Press.								

2. <https://www.abe.pl/en/book/9780191527999/smart-structures>

1. MOOC Course

1. <https://nptel.ac.in/courses/101105090>

2. <https://nptel.ac.in/courses/112104173>

3. <https://nptel.ac.in/courses/112104251>

Course Outcomes : Students should be able to:

At the end of the course the students are able to

1. Exemplify the classification, advantages and applications of smart structures used in aircraft.
2. Exemplify the material properties and Constitutive Modeling of Piezoelectric Material.
3. Outline the static and dynamic analysis of smart structures used in aircraft.
4. Outline the properties of shape memory alloys and carry-out the analysis of shape memory alloys used in aircraft.
5. Demonstrate the structural health monitoring of an aircraft structure